SOLVATHON 5.0

TourSafe

Smart Real Time Tourist Safety Monitoring & Emergency Response System using AI, Geo-fencing and Blockchain-based digital id

PROBLEM STATEMENT

01

Reactive Nature of Current Systems

Existing safety nets like travel insurance are purely reactive, compensating for losses only after a crisis has occurred.

02

Response Delays

Emergency response times in remote areas often exceed 40 minutes, which is critical in life-threatening situations

03

Impact on Tourism

Frame the problems effectively as it will set the stage of your entire pitch.

04

Identity Theft & Loss

Traditional paper IDs are easily lost or stolen in remote areas, leaving tourists stranded.

PROPOSED SOLUTION AND FEATURES

TourSafe is an AI and Blockchain-powered ecosystem that provides proactive safety through real-time anomaly detection, smart geo-fencing, and tamper-proof Digital IDs. It bridges the critical emergency response gap by integrating automated SOS triggers with a centralized authority dashboard for instantaneous rescue and e-FIR filing.

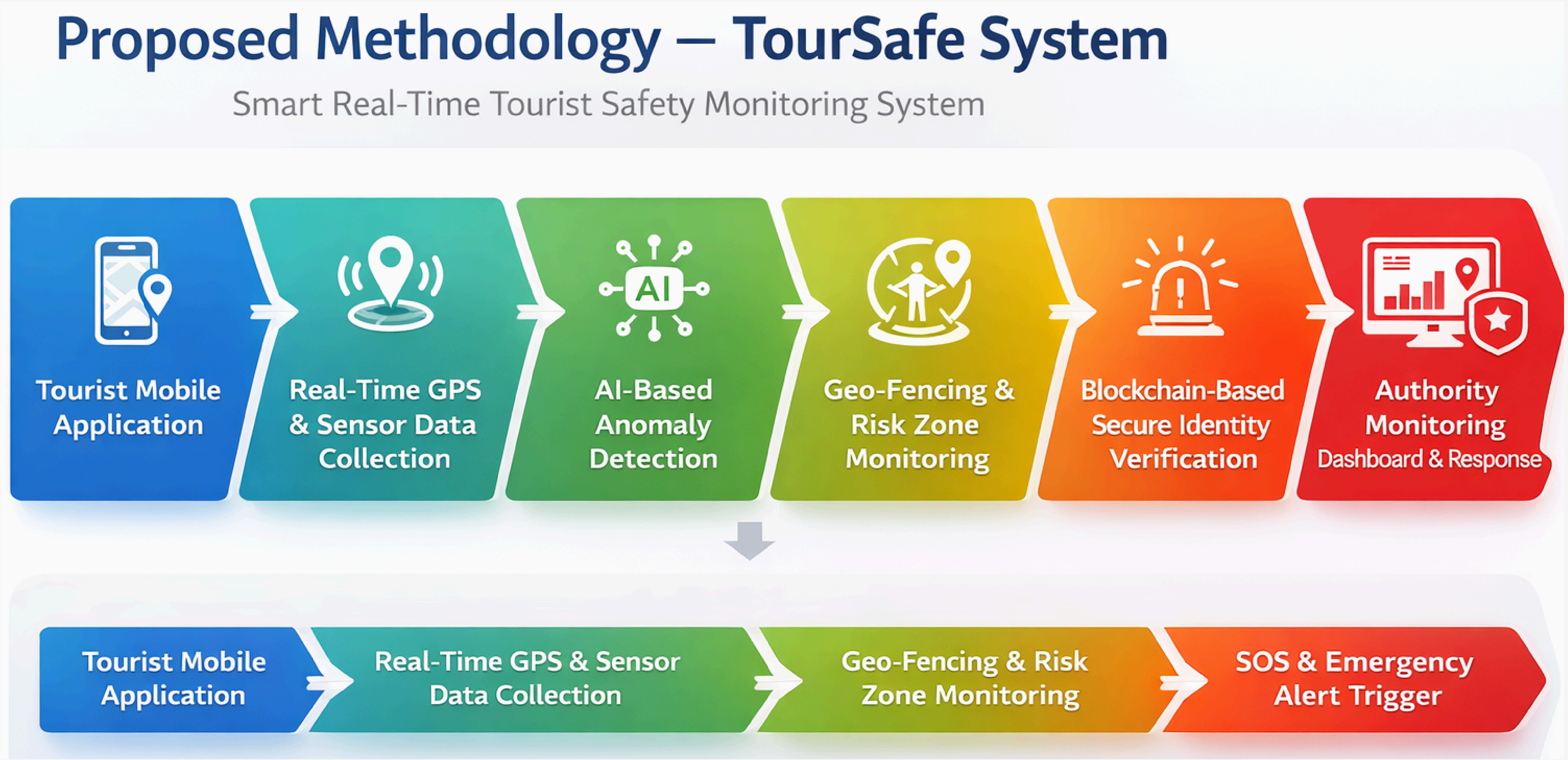
Blockchain-based Digital ID

Geo-fencing alerts for restricted zones

SOS emergency features

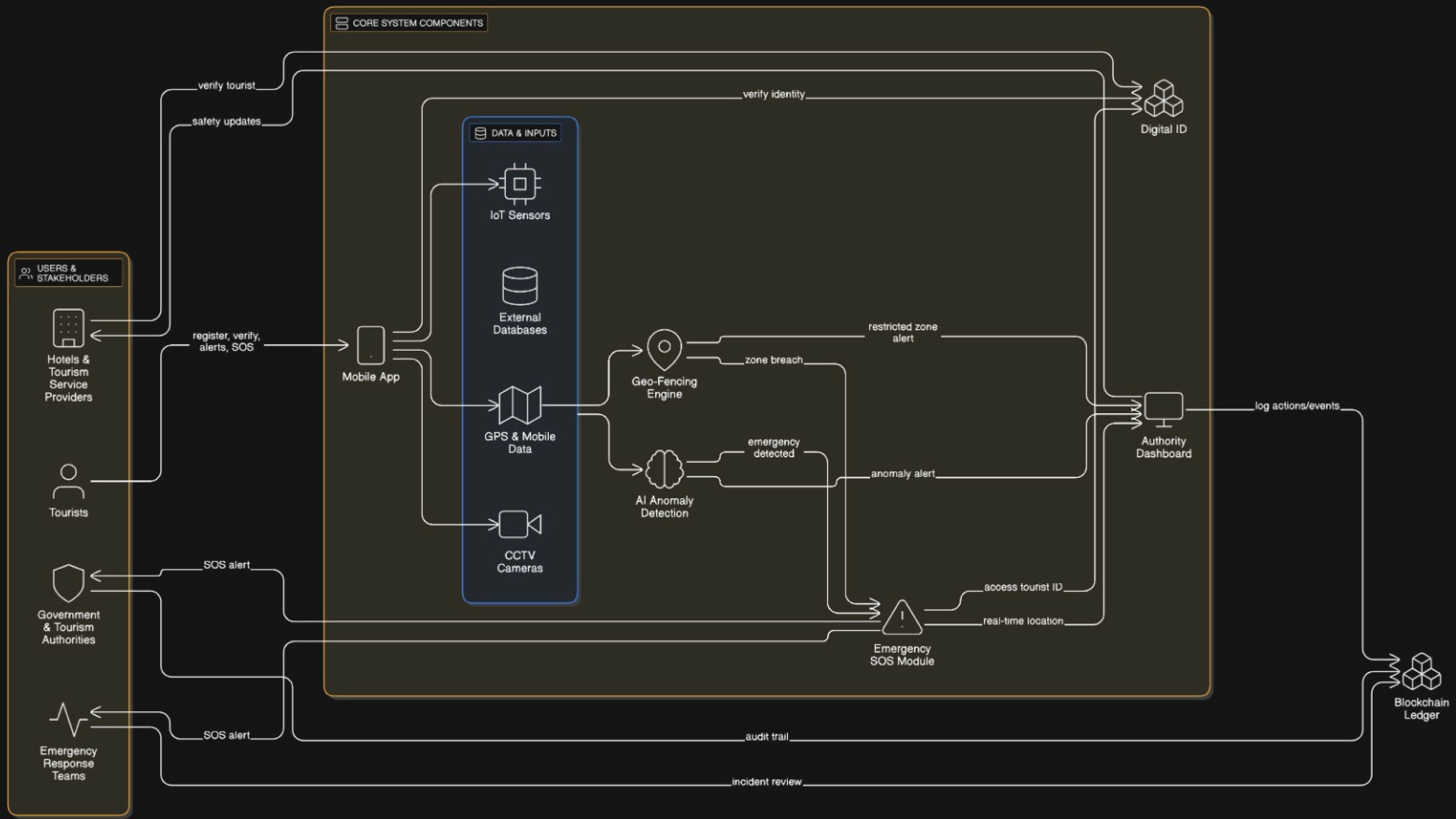
AI anomaly dashboard for unusual behavior

PROPOSED METHODOLOGY



WORKFLOW


Tourist Registration via Mobile App
Real-Time GPS Tracking & Data Collection
On-Device AI Analysis (Anomaly Detection + Geo-Fencing Check)
Secure Data Upload (Blockchain-based Digital ID + Encrypted Storage)
Alert Generation (SOS / Geo-Fence Violation / Risk Prediction)
Authority Dashboard Monitoring & Response



CORE SYSTEM COMPONENTS
USERS & STAKEHOLDERS
Hotels & Tourism Service Providers
Tourists
Government & Tourism Authorities
Emergency Response Teams
verify tourist
safety updates
register, verify, alerts, SOS
SOS alert
SOS alert
Mobile App
DATA & INPUTS
IoT Sensors
External Databases
GPS & Mobile Data
CCTV Cameras
Geo-Fencing Engine
AI Anomaly Detection
Emergency SOS Module
verify identity
restricted zone alert
zone breach
emergency detected
anomaly alert
access tourist ID
real-time location
Digital ID
Authority Dashboard
log actions/events
Blockchain Ledger
audit trail
incident review

RESEARCH AND REFERENCES

Building Travel Agent Partnerships in Tourism Industry Using Blockchain Technology to Increase Customer Satisfaction(Inayatulloh; Ahmad Fathurrozi; Wisnu Prio Pamungkas; Fried Sinlae; Wowon Priatna; Tyastuti Sri Lestari) https://ieeexplore.ieee.org/document/10767495

Anomaly detection in trajectory data for surveillance applications(Laxhammar, Rikard, Örebro University, School of Science and Technology. https://www.diva-portal.org/smash/record.jsf?dswid=1325&pid=diva2%3A440646

Smart Geo-fencing with Location Sensitive Product Affinity(SIGSPATIAL '17: Proceedings of the 25th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems) https://dl.acm.org/doi/10.1145/3139958.3140059

Automated Real-time Anomaly Detection in Human Trajectories using Sequence to Sequence Networks(Proc. IEEE Int. Conf. on Advanced Video and Signal-Based Surveillance (AVSS) pp. 1-8 (2019)) https://doi.org/10.1109/AVSS.2019.8909844

RELEVENT PATENTS

US Patent: “Interlinked geo-fencing” — US11800314B2
https://patents.google.com/patent/US11800314B2/en
WO2018126029A2 — “Blockchains for securing IoT devices” (Google Patents / WIPO)
https://patents.google.com/patent/WO2018126029 A2/en

REAL WORLD CASE STUDIES

Tourist missing / falling into icy river in Ladakh (Dec 2024) — Deccan Herald / Voice of Ladakh reports. Why: remote terrain + cold water = time-critical rescue; real-time location + anomaly detection could reduce search time.
https://voiceofladakh.in/2024/12/tourist-missing-after-falling-through-frozen-river-ice-in-ladakh/